

Adarsh Bajpai

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Summary

Software Developer shipping full-stack applications with AI-powered features built in, not bolted on. Builds backend systems, APIs, and RAG/LLM integrations that go from model to production without losing speed or reliability.

Experience

AI Engineer Intern

Jan 2026 – Present

Global Infoventures Pvt. Ltd.

Delhi, India

- Engineered an automated attendance marking system end to end, from computer vision-based detection to backend integration, achieving **~98% recognition accuracy** in pilot deployment.
- Cut retrieval-miss rate by **~40%** on a bilingual (Hindi/English) document QA system by designing and shipping the **RAG pipeline's** chunking and retrieval (top-K and heuristic ranking) layers.
- Shipped backend and content infrastructure for a **TCS joint-venture LMS** platform serving **2,00,000+ students**, contributing production-facing components at scale.

Projects

Clarity – RAG-Based Guidance Platform [\[GitHub\]](#) [\[Live App\]](#) | FastAPI, React, FAISS, RAG, LLMs, TypeScript

- Architected and shipped a full-stack multilingual RAG platform (**11 regional languages**) end to end, with an async **FastAPI + SSE streaming** backend, React frontend, and **FAISS-based retrieval** layer, live in production with a **Supabase**-hosted database.
- Engineered backend scalability improvements, including **background warmup orchestration** and frontend-offloaded static data, cutting average response latency by **~40%**.

Zen – Gamified Productivity & Habit Tracker [\[GitHub\]](#) | React, Node.js, Three.js, APIs

- Built a full-stack productivity web app unifying habit tracking, task management, finance tracking, Pomodoro timer, and note-taking into a single gamified workflow.
- Implemented an interactive **3D garden visualization** using **Three.js** that reflects user progress in real time, designed for client-side rendering to minimize server load.
- Architected the app for client-side computation, reducing backend dependency and enabling responsive, real-time interaction without server round-trips for core visualization logic.

Real-Time Multi-Object Tracking Pipeline [\[GitHub\]](#) | RT-DETR, ByteTrack, OpenCV, Pandas, PyYAML

- Built an end-to-end backend pipeline using **RT-DETR-R50** and **ByteTrack** for Indian road environments, achieving **92% tracking accuracy (MOTA)** with stable persistent object IDs across video streams.
- Processed **50+ hours of video** in batch mode by architecting a fully YAML-configurable pipeline with structured CSV logging and annotated output generation.

Leaf Disease Detection System [\[GitHub\]](#) | Python, U2-Net, ResNet, Flask, React, OpenCV, NLP

- Built a hybrid deep learning model (**U2-Net + ResNet**) trained on the **PlantVillage dataset (~60,000 images)**, achieving **96% classification accuracy**, and engineered the full-stack delivery layer (React + Flask) around it.
- Added a bilingual, **LLM-powered chatbot** interface using API calling for real-time farmer queries, integrating model inference with a production-facing web app including image upload and severity scoring.

Technical Skills

Programming: Python, JavaScript, TypeScript

Web & Backend: React, Node.js, FastAPI, Flask, Three.js, REST APIs, MongoDB

DevOps & Tools: Git, GitHub, Docker, Jenkins, AWS, Postman, Jira

AI/ML & GenAI: RAG Pipelines, LLM Integration, Computer Vision, NLP, PyTorch, Hugging Face, Vector Databases, Prompt Engineering

Data & Libraries: NumPy, Pandas, Matplotlib, OpenCV

Design & Other: Figma, Wireframing, UI Designing, Prototyping, Adobe Suite

Education

Bachelor of Technology in Computer Science

Aug. 2022 – May 2026

NIIT University | AI and ML Specialisation

Neemrana, Rajasthan

- Coursework: Machine Learning, Generative AI, AI Agents and Systems, Software Engineering, DBMS

Certifications

Oracle Cloud Infrastructure 2025 Generative AI Professional [\[Certificate\]](#)

Oracle | Generative AI, model deployment on OCI, prompt engineering, enterprise AI integration

React Web Developer Certificate [\[Certificate\]](#)

Infosys | Full-stack React development, backend APIs, routing, state management, production-ready web apps